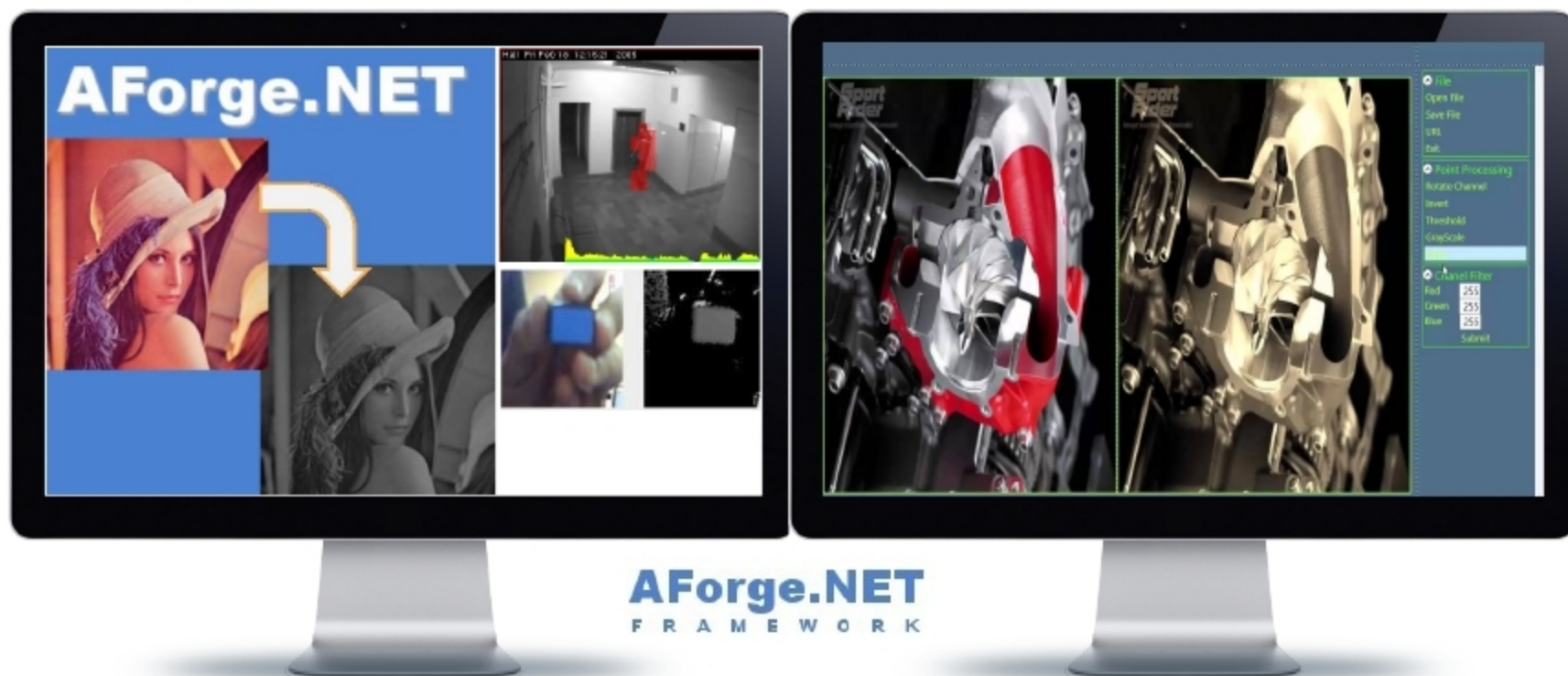


AForge.NET: A Powerful Open Source Computer Vision Tool

AForge.NET is a C# framework designed for developers and researchers in the fields of computer vision and artificial intelligence, which include image processing, neural networks, genetic algorithms, machine learning, robotics, etc. This article gives a quick introduction to this powerful framework.



Computer vision (CV) is the capability of a machine to get and analyse visual information. It comprises videos and photos, and consists of pictures from varied thermal cameras, sensors, and other different sources. CV is a branch of computer science, and is termed as a process that involves capturing, processing and analysing real world images to allow machines to extract meaningful, contextual information from the physical world. Nowadays, computer vision is the foundation and a key means of testing and exploiting deep learning models and using other artificial intelligence techniques for implementing useful and practical applications.

Since 2012, the concepts of 'context, attention and intention' are developing computer vision into a new sub-discipline of AI. Computer vision helps robots, autonomous vehicles, and everything from smart agriculture to drones to become more productive. The market for computer vision

is developing nearly as fast as the capacities are increasing. It is estimated that the global computer vision market can reach US\$ 26.2 billion by 2025. Computer vision adds a natural way for users to interact with the digital and physical world around them. It is also enabling new ways for devices to interact with users and the environment around them.

Currently, lots of commercial and open source tools are available for developers and industry R&D professionals to design applications based on computer vision. In this article, we primarily focus on the AForge.NET library.

An introduction to AForge.NET library

AForge.NET is a powerful AI and computer vision library designed by Andrew Kirillov for the .NET framework. The framework, at present, consists of many parts—image processing, neural networks, evolution algorithms, fuzzy logic, machine learning, robotics, etc.